

Submit the following five exercises. You are expected to provide full solutions in a linear and coherent fashion. You are encouraged to work in groups, but your work must be your own and must be self-contained.

Exercise 1.

Justify the following equality using a graph, a triangle, or the unit circle.

$$\sin\left(\frac{11\pi}{3}\right) = -\frac{\sqrt{3}}{2}.$$

Exercise 2.

Find all $x \in [0, 2\pi)$ satisfying $\cos(x) - \sin(x) = 0$. Make sure to fully justify your answer.

Exercise 3.

Find all $x \in [0, 2\pi)$ satisfying $2\sqrt{2}\sin(x)^2 + (\sqrt{2} + 2)\sin(x) + 1 = 0$. Make sure to fully justify your answer.

Exercise 4.

Consider the functions

$$g(x) = 2\sin(x) \quad \text{and} \quad f(x) = \begin{cases} e^{3x} & \text{if } x < 0, \\ x^3 + 2x & \text{if } 0 \leq x \leq 4, \\ \sqrt{x^2 - 1} & \text{if } 4 < x. \end{cases}$$

Determine the domain and range of the function $(f \circ g)(x) = f(g(x))$.

Exercise 5.

Find all real roots (zeros) of

$$h(x) = \cos\left(\frac{1}{x}\right).$$